SM-HAD: Spectrum Mamba for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) is vital in remote sensing and surveillance, where the goal is to identify rare or unknown targets without prior labels. However, HAD remains challenging due to: 1) high spectral dimensionality; 2) complex spectral–spatial dependencies; 3) diverse anomaly scales; and 4) the need for computationally efficient inference. To address these challenges, we propose spectrum-Mamba for HAD (SM-HAD), a unified encoder–decoder framework that integrates three complementary modules. First, the ortho spectrum Fourier block (OSFB) mitigates spectral redundancy by projecting features into the frequency domain and selectively enhancing informative spectral modes. Second, the masked vanilla attention block (MVAB) explicitly preserves local spatial relations through a context-aware masking mechanism, improving fine-grained anomaly localization. Third, the residual Mamba block (RMB) leverages state–space modeling to capture long-range spatial dependencies with linear complexity, enabling scalable inference. Extensive experiments and ablation studies on six benchmark datasets demonstrate that SM-HAD consistently outperforms 18 state-of-the-art (SoTA) methods, achieving competitive accuracy while requiring fewer parameters and lower FLOPs. These properties make SM-HAD highly suitable for real-time and resource-constrained applications. The source code is available at https://github.com/Tanishq251/SM-HAD**

***Index Terms*—Anomaly map, hyperspectral anomaly detection (HAD), masked attention, ortho spectrum Fourier block (OSFB), reconstruction-based detection, remote sensing, residual mamba, spectral–spatial modeling, spectrum-Mamba for HAD (SM-HAD), state–space model (SSM).**

I. Introduction

ANOMALY detection in hyperspectral imagery (HSI) refers to the detection of pixels or regions that significantly depart from one or more local backgrounds without prior knowledge of the anomalies present in the image [1]. Anomaly detection systems can capitalize on spectral, spatial, or joint spectral–spatial information about real or hidden targets [2]. Such anomaly detection in HSI is a formidable tool when considering applications such as environmental monitoring [3], defense intelligence [4], disaster relief and rescue [5], resource mapping [6], etc. On the way ahead, technology will make HSI anomaly detection's rightful position on the cutting edge of strategic decision-making, sustainable development, and national resilience.

Traditional hyperspectral anomaly detection (HAD) techniques generally follow a two-step pipeline: 1) background modeling [7] and 2) anomaly detection via measuring deviations from the modeled background [8]. Existing methods can be broadly categorized into statistical-based, representation-based, and deep learning-based approaches. Statistical approaches [7], [9] attempt to detect anomalies based on distribution deviations but suffer from poor adaptability to heterogeneous backgrounds [10], [11]. Representation-based methods employ sparse or low-rank subspace modeling [12], [13] but often fail to capture spectral–spatial coupling effectively. Deep learning-based methods [14], [15], [16], [17], [18] have recently demonstrated strong performance in HAD; however, they still face several critical challenges. First, most existing architectures struggle to jointly model global spectral dependencies and fine-grained spatial structures [19], [20], [21], which limits their ability to detect subtle anomalies that exhibit both spectral uniqueness and spatial coherence. Second, attention-based approaches can capture spectral–spatial relations but often suffer from high computational cost and over-smoothing effects, weakening boundary accuracy and obscuring small or context-dependent anomalies. Third, recent state–space and transformer-based models can model long-range dependencies but frequently overlook locality preservation and spectral redundancy reduction, both of which are essential for accurate anomaly localization and computational efficiency. Moreover, as shown in Fig. 1, many existing HAD models achieve strong detection accuracy but at the expense of high parameter count and computational complexity, making them less practical for real-time or resource-constrained deployment. These limitations highlight the need for a unified framework that effectively integrates spectral diversity preservation, spatial locality modeling, and scalable long-range dependency learning.

Received 17 December 2025; revised 29 January 2026; accepted 2 March 2026. Date of publication 23 March 2026; date of current version 10 April 2026. *(Tanishq Rachamalla and Aryan Das contributed equally to this work.) (Corresponding author: Lorenzo Bruzzone.)*

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Digital Object Identifier 10.1109/TGRS.2026.3676658

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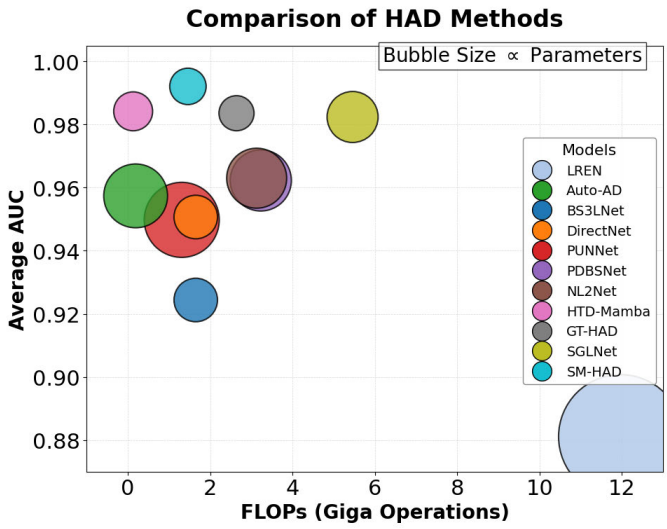


Fig. 1. Bubble chart showing the tradeoff between accuracy and FLOPs for HAD models. Bubble size indicates parameter count. SM-HAD offers the best accuracy with low computation and compact size.

Unlike CNN-based HAD methods that mainly capture local spatial patterns, attention-based models that characterize global dependencies but suffer from high computational cost and over-smoothing, and recent state–space models (SSMs) [22], [23] that model long-range sequences with linear complexity but lack spatial locality preservation, SM-HAD uniquely combines Fourier spectral decomposition, locality-aware masked attention, and linear state–space modeling in a unified encoder–decoder framework. This joint design enables efficient and accurate spectral–spatial–sequential anomaly representation.

In brief, the key contributions made in this work are as follows.

1) We propose SM-HAD, the first unified hybrid framework that integrates Fourier-based spectral modeling, locality-aware masked attention, and linear state–space dependency learning for HAD.
2) We develop the ortho spectrum Fourier block (OSFB) to capture orthogonal spectral representations using frequency-domain decomposition and selective complex filtering, enhancing discriminative spectral cues.
3) We introduce the Masked Vanilla Attention Block (MVAB), which preserves neighborhood-level spatial structure using binary locality-constrained masking, reducing oversmoothing and improving anomaly localization.
4) We propose the residual Mamba block (RMB) that incorporates linear-time state–space modeling to extract long-range spatial dependencies with high scalability and stable training dynamics.
5) Extensive experiments on six benchmark datasets demonstrate that SM-HAD outperforms 18 state-of-the-art (SoTA) methods in detection accuracy while maintaining lower FLOPs and fewer parameters, highlighting its efficiency and deployment suitability.

The remainder of this article is structured as follows. Section II reviews the related works in HAD. Section III presents the proposed SM-HAD method. Section IV describes the experimental setup. Section V presents the experimental results, comparison, ablation study, and model complexity. Section VI provides the concluding remarks.

II. Literature Summary

A. Statistics-Based Techniques

Statistical methods, with their strong theoretical base, are widely adopted in many applications of anomaly detection in three-dimensional hyperspectral image cube (3D-HSI), remote sensing, and other high-dimensional data [24]. It is assumed that background data points are generated according to some known statistical distribution, and anything that significantly deviates from the aforementioned statistical distribution of the normal ones is treated as an anomaly. Therefore, they suit the highly multivariate data that are characteristic of 3D-HSI cubes well. Set theory, including the Chi-Square Test [25] and t-tests [26], is employed to statistically assess if an observed hyperspectral sample belongs to the assumed population behavior. In addition, numerous specialized statistical and machine learning techniques have been developed to address more advanced anomaly-detection scenarios beyond classical methods. The Reed-Xiaoli Detector (RXD) [7] models the hyperspectral background as a multivariate Gaussian distribution and computes the Mahalanobis distance for each pixel to detect outliers in the 3D-HSI. Other variants, such as the weighted RXD (WRX) [9], have improved RXD by allocating a greater weight to the spatial bands most useful for improving sensitivity in a complex environment. The subspace RXD (SRX) [27] applies dimensionality reduction to RXD to improve computational efficiency and robustness against noise. The generalized likelihood ratio test (GLRT) [10] enhances detection performance by comparing the likelihoods under normal and abnormal hypotheses without requiring full knowledge of the background distribution, thereby providing the necessary flexibility across different 3D-HSI datasets. In the real-time applications of hyperspectral monitoring, distributed learning and processing with one-class SVMs [28] have been coupled together to deliver a scalable and continuously running anomaly detection. The kernel isolation forest detection (KFID) [29] merges kernel tricks with isolation forests to discover anomalies in nonlinearly separable high-dimensional 3D-HSI data, where standard linear models usually fail. Lastly, MsRFQFT [30] implements random Fourier mappings on varying scales, coupled with quasi-quadratic testing, to greatly escalate sensitivity and speed for anomaly detection over complex hyperspectral scenarios. In summary, methods ranging from statistical hypothesis testing to machine learning-integrated techniques are still the backbone of current anomaly detection pipelines in 3D-HSI, due to their mathematical rigor, generalizability, and flexibility toward diverse complexities of problems.

B. Representation-Based

Representation-based techniques on HSI data, Collaborative Representation Detection (CRD) [31] transform the 3-D hyperspectral cube into a 2-D matrix and normalize the spectrum.

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Then it represents each test pixel as a linear combination of all training pixels, such that normal pixels have small reconstruction errors, whereas anomalies have poor reconstruction capability due to their deviation from the normal subspace. Sparse CRD (SCRD) [32] generalizes CRD by adding sparsity constraints in the representation process, such that only the most informative training pixels are utilized. This adds robustness to noise and clutter naturally occurring in 3D-HSI scenes. Weighted CRD (WCRD) [33] further enhances the process by using dynamic weights on training pixels based on spectral similarity, such that the model can highlight contributions from similar pixels and suppress the effect of noisy or irrelevant pixels. This is especially useful in complex hyperspectral environments. Low-rank and sparse representation (LRASR) [34] represents the hyperspectral data as lying mainly on a low-dimensional manifold, factorizing the data into a low-rank matrix of normal patterns and a sparse matrix of anomalies, which aids in modeling structured backgrounds in 3D-HSI. Low-rank CRD (LRCRD) [35] integrates the collaborative linear reconstruction of CRD with the low-rank constraints of LRASR, modeling the relationships among training samples and the anomalies jointly while separating the anomalies with the sparse residual errors. Graph and total variation regularized low-rank representation (GTVLRR) [36] builds a graph from the 3D-HSI pixels where the nodes are the spectral signatures and the edges represent neighborhood similarities, imposing smoothness across spatially related pixels with a low-rank structure, thus improving detection performance in spatially correlated scenes. To extend its applicability to multidimensional tensors, PCA-based tensor low-rank and sparse representation (PCA-TLRSR) [13] factorize the 3D-HSI data along both spatial and spectral axes, improving the detection of minute anomalies from very high-dimensional and richly structured hyperspectral cubes. Finally, learnable background endmember with subspace representation (LEBSR) [37] is specifically tailored for hyperspectral applications. It learns a set of background endmembers from the 3D-HSI, represents the background using a low-dimensional subspace, and detects anomalies as residuals from this learned model, providing accurate performance in complex hyperspectral settings with mixed pixels.

C. Deep Learning-Based Techniques

Deep learning has gained substantial traction in anomaly detection because it eliminates the need for manual parameter tuning or prior distributional assumptions, instead identifying anomalies through high reconstruction errors in self-supervised models. Among these approaches, Auto-AD [16] employs a fully convolutional autoencoder to reconstruct the original hyperspectral cube from random noise input, with anomalies manifesting as large reconstruction errors. WeaklyAD [17] leverages a spectral-structured generative adversarial network (GAN) to reconstruct the 3D-HSI; abnormal pixels consistently exhibit higher reconstruction errors than the normal background, enabling weakly supervised learning.

Departing from reconstruction-based methods, deep support vector data description (DSVDD) [18] formulates anomaly detection as a classification task by mapping deep spectral–spatial features onto a compact hypersphere, where any feature lying outside the hypersphere during testing is flagged as anomalous. Hybrid models such as the low-rank embedded network (LREN) [19] combine an autoencoder with a Gaussian mixture model (GMM) to jointly learn robust spectral features, while anomaly detection is subsequently performed using low-rank representation-based techniques applied to the learned deep features. More recent advancements, such as anomaly detection using masked autoencoders (ADMAEs) [20], reconstruct masked spectral–spatial tokens during pretraining to capture normal data patterns, with anomalies inferred through inconsistencies in reconstruction. SMMs have recently emerged as efficient alternatives to transformers for sequence modeling. Mamba [22] achieves linear-time complexity $\mathcal{O}(L)$ through selective state–space mechanisms, significantly reducing the quadratic cost of attention. The selective scanning mechanism enables content-aware filtering by dynamically adjusting state transitions based on input tokens, allowing the model to focus on relevant information while discarding irrelevant context. Recent work [38] has unified transformers and SSMs under a structured state–space duality framework, demonstrating that attention can be viewed as a special case of structured SSMs. VMamba [23] extended this approach to vision tasks by introducing cross-scan mechanisms that traverse 2-D spatial feature maps in multiple directions, demonstrating competitive performance on image classification and segmentation tasks. While SSMs excel at modeling long-range dependencies with linear complexity, their application to HAD remains limited, motivating the integration of state–space modeling with explicit spectral and local spatial processing in SM-HAD.

Recent advances in HAD have increasingly focused on enhancing background reconstruction, spatial–spectral modeling, and efficient sequence learning. BockNet [39] employs a blind-block strategy with a guard window to strictly isolate anomalous pixels during background learning, preventing anomaly leakage. Similarly, PUNNet [40] integrates patch-shuffle downsampling with a nonlinear activation-free (NAFNet) block and dilated convolutions in a blind-spot architecture, ensuring that the receptive field excludes the center pixel. This improves global contextual modeling while suppressing anomaly information during reconstruction, thereby increasing anomaly reconstruction errors. PDBSNet [41] introduces a pixel-shuffle downsampling strategy combined with a blind-spot network that processes the entire hyperspectral image directly without patch cropping, enabling efficient full-image background reconstruction while maintaining the blind-spot constraint through its downsampling mechanism. DirectNet [42] introduces a dual-window blind-block architecture in which the central region is masked to form an inner window that is invisible during reconstruction. The network relies solely on outer-window information to reconstruct the center, forcing the output to align with the background distribution and amplifying reconstruction errors for anomalies. FTSGAN [43] utilizes a frequency-to-spectrum mapping mechanism for semi-supervised detection, effectively learning to map frequency components to spectral signa-

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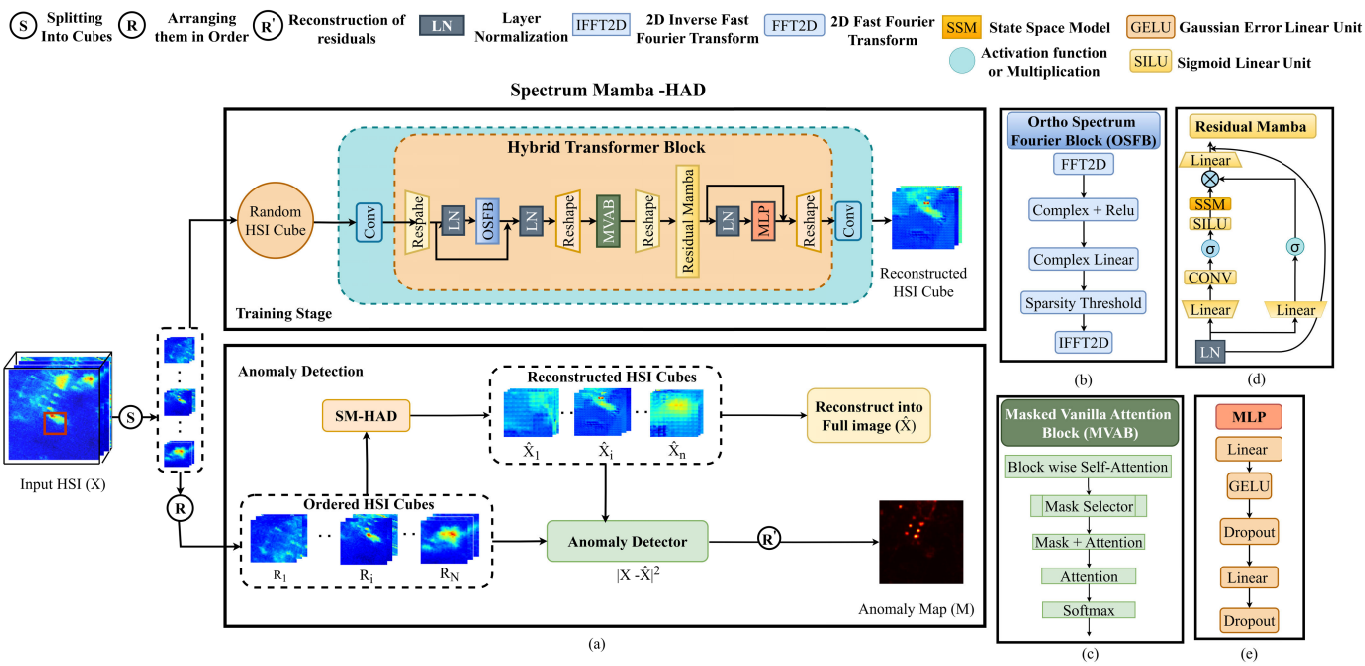


Fig. 2. Architecture of SM-HAD illustrating (a) training (top) and anomaly detection (bottom) pipeline. Submodules represent (b) OSFB, (c) Residual Mamba, (d) MLP, and (e) MVAB, respectively.

tures. Additionally, a simple variational joint discrimination network [44] has been proposed to address cross-image HAD, improving generalization across different scenes. NL2Net [45] enhances nonlocal spatial–spectral modeling by capturing relationships between distant pixels and spectral bands, improving background–anomaly separability in heterogeneous scenes. BS3LNet [46] adopts a blind-spot self-supervised strategy to reconstruct only background pixels while excluding the pixel itself from the receptive field. This design prevents anomaly information leakage, improves spectral–spatial feature representation, and enhances anomaly detection based on reconstruction errors. GT-HAD [21] introduces a gated transformer architecture that explicitly exploits spatial–spectral similarity for anomaly discrimination. It employs a dual-branch network, where one branch models background features and the other focuses on anomaly responses. A content-matching mechanism (CMM) and an adaptive gating unit dynamically regulate branch activation, strengthening anomaly localization and reducing false positives. HTD-Mamba [47] employs pyramid state–space modeling with spectrally contrastive learning to efficiently capture long-range spectral dependencies with linear complexity. Although originally designed for hyperspectral target detection with known spectral signatures, it can be adapted for anomaly detection by removing target prior requirements and using residual anomaly scoring. SGLNet [48] addresses the insufficient integration of feature hierarchies by employing a three-branch architecture that simultaneously extracts global, local, and spectral low-rank features. It utilizes a graph convolution module to capture global background dependencies and a memory matrix to explicitly model intrinsic low-rank spectral priors. These features are then integrated via a spectral-guided feature fusion module (SFFM) to dynamically adjust weights and reduce redundancy before background prediction. Overall, while existing methods address specific aspects such as blind-spot reconstruction or long-range modeling, most approaches focus on isolated components. They often overlook the seamless integration of frequency-domain analysis, sequential dependencies, and local spatial structures within a unified framework, which is the primary focus of our proposed SM-HAD.

III. Methodology

A. Overview

The proposed architecture, spectrum-Mamba for HAD (SM-HAD), is designed to integrate spectral processing, masked self-attention, and efficient sequence modeling to capture diverse contextual dependencies in spectral–spatial data. As shown in Fig. 2(a), the architecture comprises three core stages: an input embedding layer, a hybrid Transformer block, and an output projection layer. The overall embedding flow is structured to preserve spatial hierarchies while enabling multidomain feature extraction.

B. Input Embedding

Let $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ be an HSI image, where C is the number of spectral channels and H, W are the spatial dimensions. The cube is divided into N overlapping blocks $\hat{\mathbf{X}}_i \in \mathbb{R}^{P\times P\times C}$ of size $P \times P \times C$ using a stride of s pixels, where $i = 1, 2, \ldots, N$.

The proposed network starts with a convolutional embedding layer of kernel size 3×3 to project each patch into a latent space of dimension D

$$\mathbf{Z}_0^{(i)} = \mathrm{Conv}_{3\times 3}\left(\hat{\mathbf{X}}_i\right) \in \mathbb{R}^{D\times P\times P}, \quad i = 1, \ldots, N. \tag{1}$$

Each embedded patch $\mathbf{Z}_0^{(i)}$ is then permuted to a channel-last format for subsequent processing

$$\mathbf{Z}_0^{(i)} \in \mathbb{R}^{P\times P\times D}, \quad i = 1, \ldots, N. \tag{2}$$

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C. Hybrid Transformer Block

As shown in Fig. 2(a), the proposed hybrid Transformer block consists of four core components: 1) an OSFB; 2) a MVAB; 3) a RMB; and 4) a multilayer perceptron (MLP). The OSFB primarily leverages frequency-domain transformations to efficiently capture global spectral correlations across hundreds of spectral bands, naturally aligning with the inherent spectral correlation patterns in HSI data [Fig. 2(b)]. The MVAB employs strategic masking to prevent over-smoothing of fine-grained spatial feature which is critical for preserving subtle anomaly signatures in high-resolution HSI [Fig. 2(c)]. The RMB provides linear computational complexity for modeling of long-range spatial dependencies while ensuring stable gradient flowthrough deep architectures processing high-dimensional spectral information [Fig. 2(d)]. Finally, the MLP enhances the nonlinear feature transformation capabilities for distinguishing subtle anomalies from normal background variations in hyperspectral scenes [Fig. 2(e)]. The detailed description of each individual component is given as follows.

1) Ortho Spectrum Fourier Block: The OSFB is designed to capture global spectral dependencies through frequency-domain processing, which is crucial for the detection of hyperspectral anomalies. To achieve this, the block processes the input features using a sequence of normalization, spectral transformation, learnable complex filtering, sparsity enhancement, and residual integration to obtain robust and context-aware representations.

Let $\mathbf{Z}_0^{(i)} \in \mathbb{R}^{P\times P\times D}$ be ith embedded patch. For simplicity of notation, we drop the patch index superscript (i) in the following equations and refer to a single patch as $\mathbf{Z}_0$. The feature tensor is first normalized across its feature channels to stabilize the distribution and facilitate effective frequency-domain learning

$$\hat{\mathbf{Z}}_0 = \text{LayerNorm}\left(\mathbf{Z}_0\right). \tag{3}$$

Then, the normalized features are projected into the frequency domain by applying a 2-D Fast Fourier Transform (FFT2D) across the spatial dimensions of each channel, as shown in Fig. 2(b)

$$\hat{\mathbf{Z}} = \text{FFT2D}\left(\hat{\mathbf{Z}}_0\right) \tag{4}$$

which decomposes the spatial feature map into its constituent frequency components, forming a complex-valued tensor with real and imaginary parts associated with each spatial–spectral element. We note that the FFT itself projects inputs onto an orthogonal basis of complex exponentials, while the subsequent learned complex filters and nonlinearities operate on the resulting coefficients and therefore do not, in general, preserve strict orthogonality; despite this, the initial orthogonal decomposition reduces spectral redundancy and facilitates targeted frequency-domain filtering. This transformation exposes global spectral dependencies that are not easily accessible in the spatial domain.

In each frequency and channel dimension, two phases of complex-valued filtering operations are performed, each involving learnable real and imaginary projection matrices and bias terms. Let $\mathbf{W}_k^{\mathbb{R}}, \mathbf{W}_k^{\mathbb{I}} \in \mathbb{R}^{D\times D}$ and $\mathbf{b}_k^{\mathbb{R}}, \mathbf{b}_k^{\mathbb{I}} \in \mathbb{R}^{D}$ represent the real and imaginary parts of the learned parameters for stage $k = 1, 2$, respectively. The OSFB employs *two* such complex filtering stages because a single linear complex transformation is insufficient to model the higher-order relationships between magnitude and phase that characterize subtle hyperspectral anomalies. The cascade of a nonlinear stage followed by a linear refinement significantly enhances the expressive capacity of the overall transformation.

a) First complex filtering stage:

$$\mathbf{Z}_1^{\mathbb{R}} = \text{ReLU}\left(\mathbf{W}_1^{\mathbb{R}} \cdot \text{Re}\left(\hat{\mathbf{Z}}\right) - \mathbf{W}_1^{\mathbb{I}} \cdot \text{Im}\left(\hat{\mathbf{Z}}\right) + \mathbf{b}_1^{\mathbb{R}}\right) \tag{5}$$

$$\mathbf{Z}_1^{\mathbb{I}} = \text{ReLU}\left(\mathbf{W}_1^{\mathbb{I}} \cdot \text{Re}\left(\hat{\mathbf{Z}}\right) + \mathbf{W}_1^{\mathbb{R}} \cdot \text{Im}\left(\hat{\mathbf{Z}}\right) + \mathbf{b}_1^{\mathbb{I}}\right). \tag{6}$$

These equations correspond to the complex multiplication rule $(A + iB)(a + ib) = (Aa - Bb) + i(Ba + Ab)$, where $\mathbf{W}_1^{\mathbb{R}} + i\mathbf{W}_1^{\mathbb{I}}$ operates as a learnable complex filter. This first stage introduces a nonlinear activation into the frequency domain, which is crucial because anomalies in hyperspectral data often manifest not only through changes in magnitude but also through *subtle phase shifts*. While a purely real or magnitude-only filter could overlook such phase variations, the complex formulation explicitly models both amplitude and phase interactions. The use of ReLU here provides a strong sparsifying effect that helps emphasize informative frequency components while suppressing noise, creating a clean foundation for subsequent refinement.

b) Second complex filtering stage:

$$\mathbf{Z}_2^{\mathbb{R}} = \mathbf{W}_2^{\mathbb{R}} \cdot \mathbf{Z}_1^{\mathbb{R}} - \mathbf{W}_2^{\mathbb{I}} \cdot \mathbf{Z}_1^{\mathbb{I}} + \mathbf{b}_2^{\mathbb{R}} \tag{7}$$

$$\mathbf{Z}_2^{\mathbb{I}} = \mathbf{W}_2^{\mathbb{I}} \cdot \mathbf{Z}_1^{\mathbb{R}} + \mathbf{W}_2^{\mathbb{R}} \cdot \mathbf{Z}_1^{\mathbb{I}} + \mathbf{b}_2^{\mathbb{I}}. \tag{8}$$

The second complex filtering stage applies another complex projection to $\mathbf{Z}_1^{\mathbb{R}}$ and $\mathbf{Z}_1^{\mathbb{I}}$, thereby refining the nonlinear frequency activations produced in stage 1. This stage is intentionally *linear* to stabilize the representation while allowing the model to learn more abstract spectral interactions that a single filtering stage cannot capture. The composition of nonlinear selection (stage 1) and linear refinement (stage 2) forms a two-step complex transformation with higher expressive power, enabling the OSFB to learn multistage, phase-aware spectral descriptors essential for robust anomaly detection.

The outputs $\mathbf{Z}_2^{\mathbb{R}}$ and $\mathbf{Z}_2^{\mathbb{I}}$ represent progressively enriched frequency-domain features that support the detection of both strong and subtle anomalies distributed across the scene.

c) Aggregation, sparsification, and reconstruction: The stage-2 outputs are merged into a complex tensor

$$\hat{\mathbf{Z}}_{\text{out}} = \mathbf{Z}_2^{\mathbb{R}} + j\,\mathbf{Z}_2^{\mathbb{I}} \tag{9}$$

where $j = \sqrt{-1}$ denotes the imaginary unit.

To enhance robustness, a soft-shrinkage operation is applied

$$\hat{\mathbf{Z}}_{\text{shrink}} = \text{SoftShrink}\left(\hat{\mathbf{Z}}_{\text{out}}, \lambda\right) \tag{10}$$

where the threshold λ suppresses weak or noisy frequency coefficients and preserves structurally significant ones. This sparsity-inducing operation is a key factor in isolating anomaly-relevant responses.

An inverse FFT reconstructs the enhanced spatial representation

$$\mathbf{Z}_{\text{rec}} = \text{IFFT2D}\left(\hat{\mathbf{Z}}_{\text{shrink}}\right). \tag{11}$$

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Finally, a residual connection injects the refined spectral information back into the spatial domain

$$\mathbf{Z}_1 = \mathbf{Z}_0 + \mathrm{Re}\left(\mathbf{Z}_{\mathrm{rec}}\right) \tag{12}$$

where $\mathrm{Re}(\cdot)$ extracts the real part of the complex-valued tensor. A concluding layer normalization stabilizes the final output

$$\hat{\mathbf{Z}}_1 = \mathrm{LayerNorm}\left(\mathbf{Z}_1\right). \tag{13}$$

d) Choice of activation and empirical evidence: We use ReLU in the first complex filtering stage because its hard thresholding synergizes well with soft-shrinkage and encourages strong spectral sparsity, which benefits anomaly discrimination. Although smoother activations (such as SiLU and GELU) were evaluated, they produced only negligible changes in performance because soft-shrinkage dominates the nonlinear behavior of OSFB.

2) Masked Vanilla Attention Block: The MVAB, see Fig. 2(c), is designed to model and enhance local spatial relationships within feature maps, while adaptively restricting attention only to the contextually important neighborhoods. This prevents oversmoothing and preserves spatially localized features which are essential for HAD.

Given the output $\hat{\mathbf{Z}}_1$ obtained from the OSFB, MVAB partitions the feature map into overlapping patches of size $p_s \times p_s$ (where $p_s = 3$), extracted with stride $p_{st} = 3$. This restructuring transforms the entire 2-D feature map into a set of local and context-preserving vectors. Each patch is flattened into a 1-D vector, yielding a patch matrix $\mathbf{A}_{\mathrm{patch}} = \mathrm{FlattenPatches}(\hat{\mathbf{Z}}_1)$, whose rows represent the vectorized patches. To achieve flexible contextual modeling, the architecture uses two independent modules to convert the patches into query (q) and value (v) matrices for attention computation

$$\mathbf{q} = \mathbf{A}_{\mathrm{patch}}\mathbf{W}_q + \mathbf{b}_q, \quad \mathbf{v} = \mathbf{A}_{\mathrm{patch}}\mathbf{W}_v + \mathbf{b}_v \tag{14}$$

where $\mathbf{q}, \mathbf{v} \in \mathbb{R}^{N_{\mathrm{patch}} \times d_q}$, $\mathbf{W}_q, \mathbf{W}_v \in \mathbb{R}^{(p_s^2 D) \times d_q}$ are the projection weight matrices, $\mathbf{b}_q, \mathbf{b}_v \in \mathbb{R}^{d_q}$ are the bias vectors, D represents the number of input feature channels (i.e., the embedding dimension for each spatial location), $d_q = D/2$ is the lower-dimensional latent space, and N_{patch} is the total number of extracted patches. These projections serve to reduce the dimensionality while capturing the core local characteristics for each patch. Then the self-attention is computed among these queries to detect which patches should communicate effectively, as modeled by

$$\mathbf{A} = \frac{\mathbf{q}\mathbf{q}^T}{\sqrt{d_q}} \tag{15}$$

where $\mathbf{A} \in \mathbb{R}^{N_{\mathrm{patch}} \times N_{\mathrm{patch}}}$ quantifies patchwise similarity and allows the model to learn affinities and local dependencies across the spatial domain.

To structure and control information, MVAB constructs an attention mask $\mathbf{M}$ using a binary context matrix $\mathbf{m}$, which determines permissible attention connections between the patches. The elements of the binary context matrix $\mathbf{m}$ can be constructed as follows.

1) $\mathbf{m} \in \{0,1\}^{N_{\mathrm{patch}} \times N_{\mathrm{patch}}}$ encodes neighbor relationships between all pairs of patches.
2) For each patch pair (a, b), set $\mathbf{m}_{a,b} = 1$ if patch b should be considered a neighbor of patch a (for example, patches that overlap or are adjacent in the spatial grid), and $\mathbf{m}_{a,b} = 0$ otherwise.
3) The specific neighborhood criteria can be based on spatial proximity—such as being within a certain distance, directly adjacent, or overlapping.

a) Practical mask construction: In our implementation, patches are indexed in a row-major order following the raster scan of the 2-D feature map. To ensure numerical stability and to avoid degenerate self-attention, the mask reduces to a diagonal self-suppression mechanism in practice: $\mathbf{M}_{a,a} = -\infty$ and $\mathbf{M}_{a,b} = 0$ for $a \neq b$. This formulation prevents a patch from attending to itself while allowing unrestricted contextual interactions across all other patch tokens. Importantly, this masking strategy is fully consistent with the extraction parameters (p_s, p_{st}), since both the attention computation and the subsequent reconstruction rely on the same spatial layout and overlapping pattern.

The attention mask is then defined as

$$\mathbf{M}_{a,b} = \begin{cases} -\infty, & a = b \\ 0, & \mathbf{m}_{a,b} = 1 \\ -\infty, & \mathbf{m}_{a,b} = 0. \end{cases} \tag{16}$$

This mask selector allows only selected off-diagonal connections, so that only spatially or contextually relevant neighbors influence each patch. The use of $-\infty$ ensures that nonallowed self-attention scores are excluded (set to zero probability) after the softmax operation, while $\mathbf{m}_{a,b} = 1$ enables valid neighboring patches to attend to each other without restriction.

To further enhance contextual integration, the masked attention is refined with a feedback mechanism

$$\hat{\mathbf{A}} = \mathrm{Softmax}\left(\mathbf{A} + \mathbf{M} + \mathrm{Softmax}\left(\frac{\mathbf{A}\mathbf{A}^T}{\sqrt{N_{\mathrm{patch}}}}\right)\right) \tag{17}$$

where $\hat{\mathbf{A}} \in \mathbb{R}^{N_{\mathrm{patch}} \times N_{\mathrm{patch}}}$ is the adaptively weighted attention matrix. This feedback incorporates higher-order and compound patch relations, increasing the sensitivity to the subtle spatial patterns. As a result, the obtained features are aggregated as follows:

$$\mathbf{B}_{\mathrm{patch}} = \hat{\mathbf{A}}\mathbf{v} \tag{18}$$

and further reassembled to reconstruct the spatial feature map

$$\mathbf{Z}_2 = \mathrm{Reconstruct}\left(\mathbf{B}_{\mathrm{patch}}\right) \in \mathbb{R}^{P \times P \times D}. \tag{19}$$

This ensures that each output location benefits from contextually aggregated, spatially aware features.

3) Residual Mamba Block: The RMB shown Fig. 2(d) establishes long-range sequential dependencies efficiently, which is crucial to capture spatial patterns that span over the entire input. By using a structured state–space sequence model (Mamba) within a residual framework, the RMB aggregates both local and global information with scalability and stability. Given the attention-enhanced features $\mathbf{Z}_2$, layer normalization is first applied

$$\mathbf{U} = \mathrm{LayerNorm}\left(\mathbf{Z}_2\right). \tag{20}$$

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Layer normalization ensures that the subsequent computations are numerically stable and the input statistics are consistent. The normalized tensor $\mathbf{U}$ is reshaped from the spatial grid to a sequence of length $L = P \times P$, each of having dimension$D$

$$\mathbf{U}_{\text{seq}} = \text{Reshape}(\mathbf{U}) \in \mathbb{R}^{L \times D}. \tag{21}$$

The reshape follows row-major order. This reshaping operation converts the spatial feature map into a time-like series for efficient sequential modeling. A linear projection then splits each sequence token into two parallel streams

$$\left[\mathbf{U}^{(1)}, \mathbf{G}\right] = \text{Linear}\left(\mathbf{U}_{\text{seq}}\right) \tag{22}$$

where $\mathbf{U}^{(1)} \in \mathbb{R}^{L \times D}$ is the main processing stream, and $\mathbf{G} \in \mathbb{R}^{L \times D}$ is the residual modulation stream. In order to enrich the local context, a depthwise 1-D convolution is applied across$\mathbf{U}^{(1)}$

$$\mathbf{U}^{(2)} = \text{Conv1D}\left(\mathbf{U}^{(1)}\right) \in \mathbb{R}^{L \times D}. \tag{23}$$

This mixes information from neighboring tokens before the state update. A nonlinear activation (SiLU) followed by a linear projection layer produces the delta and state–space parameters

$$\left[\mathbf{\Delta}, \mathbf{B}^{\text{ssm}}, \mathbf{C}^{\text{ssm}}\right] = \text{Linear}\left(\text{SiLU}\left(\mathbf{U}^{(2)}\right)\right) \tag{24}$$

where $\mathbf{\Delta} \in \mathbb{R}^{L \times R}$ (for delta-timestep modulation, with R delta rank), and $\mathbf{B}^{\text{ssm}}, \mathbf{C}^{\text{ssm}} \in \mathbb{R}^{L \times d_s}$ (for state–space updates, with d_s state dimension). To stabilize the Delta-time projections and make the responses positive, a softplus operation is adopted as follows:

$$\boldsymbol{\delta}_t = \text{softplus}\left(\text{Linear}\left(\mathbf{\Delta}_t\right)\right) \in \mathbb{R}^{D}. \tag{25}$$

For the state–space update, For each sequential position $t \in \{1, \ldots, L\}$, the Mamba structured scan operates as

$$\mathbf{S}_t = f\left(\mathbf{S}_{t-1},\ \mathbf{U}_t^{(2)},\ \mathbf{A}^{\text{ssm}},\ \mathbf{B}_t^{\text{ssm}},\ \mathbf{C}_t^{\text{ssm}},\ \boldsymbol{\delta}_t\right) \tag{26a}$$

$$\mathbf{Y}_t = \boldsymbol{\gamma} \odot \mathbf{U}_t^{(2)} + \mathbf{S}_t + \mathbf{G}_t \tag{26b}$$

where $\mathbf{S}_{t-1} \in \mathbb{R}^{D \times d_s}$ is the previous latent state, $\mathbf{A}^{\text{ssm}} \in \mathbb{R}^{D \times d_s}$ is the learned decay/transition rates, $\mathbf{B}_t^{\text{ssm}}, \mathbf{C}_t^{\text{ssm}} \in \mathbb{R}^{d_s}$ are time-varying input/output projection parameters, $\boldsymbol{\delta}_t \in \mathbb{R}^{D}$ controls update rates, $f(\cdot)$ is the recurrent state update of Mamba, $\mathbf{Y}_t \in \mathbb{R}^{D}$ is the final output, $\boldsymbol{\gamma} \in \mathbb{R}^{D}$ is a learned channel-wise skip/scale parameter, $\odot$ denotes element-wise multiplication, and $\mathbf{G}_t$ is the residual modulation gate.

The sequence of outputs $\mathbf{Y} = [\mathbf{Y}_1, \ldots, \mathbf{Y}_L]$ is projected back to the feature dimension

$$\mathbf{U}_{\text{mamba}} = \text{Linear}\left(\mathbf{Y}\right) \in \mathbb{R}^{L \times D}. \tag{27}$$

Finally, reshaping $\mathbf{U}_{\text{mamba}}$ to the spatial grid and applying the residual connection yields the final output $\mathbf{Z}_3$

$$\mathbf{Z}_3 = \mathbf{Z}_2 + \text{Reshape}\left(\mathbf{U}_{\text{mamba}}\right). \tag{28}$$

This residual path ensures training stability and leverages both local and long-range spatial dependencies through the Mamba SMM.

Algorithm 1 SM-HAD Training Procedure

Require: Hyperspectral image cube $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$
Ensure: Trained parameters $\Theta_{\text{SM-HAD}}$:
$\{\mathbf{W}_{\text{conv}}^{\text{in}}, \mathbf{W}_1^{\mathbb{R}}, \mathbf{W}_1^{\mathbb{I}}, \mathbf{W}_2^{\mathbb{R}}, \mathbf{W}_2^{\mathbb{I}}, \mathbf{b}_1^{\mathbb{R}}, \mathbf{b}_1^{\mathbb{I}}, \mathbf{b}_2^{\mathbb{R}}, \mathbf{b}_2^{\mathbb{I}}, \mathbf{W}_q, \mathbf{W}_v, \mathbf{b}_q, \mathbf{b}_v, \mathbf{A}^{\text{ssm}}, \boldsymbol{\gamma}, \mathbf{W}_{\text{RMB}}, W_1, W_2, \mathbf{W}_{\text{conv}}^{\text{out}}\}$

```
Extract overlapping patches {X_i}_{i=1}^N of size P × P × C with stride s
for each epoch e = 1, ..., E do
    for each mini-batch B = {X_i} do
        Forward Pass:
        Input Embedding (Eq. (1)–(2)):
          Z_0 ∈ R^{B×P×P×D}
        OSFB (Eq. (3)–(13)):
          Z_0 --FFT, filtering, IFFT--> Z_1
        MVAB (Eq. (14)–(19)):
          Z_1 --masked attention--> Z_2
        RMB (Eq. (20)–(28)):
          Z_2 --SSM scan--> Z_3
        MLP (Eq. (29)): Z_3 → Z_4
        Output Projection (Eq. (30)):
          X̂ = Conv_{3×3}(Permute(Z_4))
          ∈ R^{B×C×P×P}
        Loss Computation (Eq. (31)):
          L_MSE = 1/(NCP^2) Σ_{i=1}^N Σ_{c=1}^C Σ_{h=1}^P Σ_{w=1}^P
          (X̂_{i,c,h,w} − X_{gt,i,c,h,w})^2
        Backward Pass: Update Θ_SM-HAD via Adam
    end for
end for
```

4) Feedforward MLP: To further enhance nonlinear feature representation, a standard two-layer feedforward network (MLP) with GELU activation and a residual connection is applied to each spatial location. As shown in Fig. 2(d), the following sequence of computation is defined on the output of RMB $\mathbf{Z}_3 \rightarrow \mathbf{Z}_4 \in \mathbb{R}^{P \times P \times D}$

$$\mathbf{Z}_4 = \mathbf{Z}_3 + W_2\left(GELU\left(W_1\left(\text{LayerNorm}\left(\mathbf{Z}_3\right)\right)\right)\right) \tag{29}$$

where the MLP consists of two pointwise 1×1 convolutions or a linear layer with the associated weights W_1 and W_2 and a GELU nonlinearity, applied independently to each position.

D. Output Projection

The final processed features $\mathbf{Z}_4 \in \mathbb{R}^{P \times P \times D}$ are permuted to channel-first format $\mathbb{R}^{D \times P \times P}$ for compatibility with the convolutional decoder. A 3×3 convolution is then applied to reconstruct the output

$$\hat{\mathbf{X}} = \text{Conv}_{3\times3}\left(\text{Permute}\left(\mathbf{Z}_4\right)\right) \in \mathbb{R}^{C \times P \times P} \tag{30}$$

where $\hat{\mathbf{X}}$ denotes the reconstructed hyperspectral patch, and C is the number of spectral channels.

E. Objective Function

The overall training procedure of the proposed SM-HAD framework, encompassing the forward pass and parameter optimization, is summarized in Algorithm 1. The network

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is trained in a self-supervised reconstruction manner, where each input patch $\mathbf{X}_i \in \mathbb{R}^{C\times P\times P}$ is used as its own ground truth. We denote the ground-truth as $\mathbf{X}_{\text{gt},i}$, and the network reconstruction as $\hat{\mathbf{X}}_i$. The mean squared error (mse) loss is used to measure the pixel-wise and spectral reconstruction discrepancy

$$\mathcal{L}_{\text{mse}} = \frac{1}{NCP^2}\sum_{i=1}^{N}\sum_{c=1}^{C}\sum_{h=1}^{P}\sum_{w=1}^{P}\left(\hat{\mathbf{X}}_{i,c,h,w} - \mathbf{X}_{\text{gt},i,c,h,w}\right)^2 \tag{31}$$

where N is the total number of patches. Minimizing this loss encourages the network to accurately reconstruct both the spectral and spatial structures of the hyperspectral data.

F. Anomaly Detection

After training, anomaly detection is performed by computing reconstruction residuals for each input patch. Given an input patch $\mathbf{X}_i$ and its reconstruction $\hat{\mathbf{X}}_{\text{recon},i}$, the residual cube is defined as

$$\mathbf{R}_i = \left(\mathbf{X}_i - \hat{\mathbf{X}}_{\text{recon},i}\right)^2 \in \mathbb{R}^{P\times P\times C} \tag{32}$$

where the square is applied element-wise, indicating pixel-wise spectral errors. To address the weakened residuals near anomaly boundaries due to receptive-field mixing, a residual diffusion (RD) postprocessing step is employed. Specifically, the residual cubes are spatially diffused using 3-D average pooling

$$\mathbf{E}_i = \text{AvgPool3d}\left(\mathbf{R}_i\right) \tag{33}$$

and the enhanced residual map for the patch is obtained by summing the channel-wise errors

$$\mathbf{m}_i^{\text{RD}} = \sum_{c=1}^{C}\mathbf{E}_i^{(c)} \in \mathbb{R}^{P\times P}. \tag{34}$$

RD increases sensitivity to subtle and boundary anomalies, improving overall detection accuracy. Finally, since patches are extracted with spatial overlap, the enhanced local residual maps $\{\mathbf{m}_i^{\text{RD}}\}_{i=1}^N$ are folded and averaged to construct the final global anomaly map

$$\mathbf{M} = F\left(\left\{\mathbf{m}_i^{\text{RD}}\right\}_{i=1}^N\right) \in \mathbb{R}^{H\times W} \tag{35}$$

where F denotes the folding and aggregation function, and $H \times W$ is the spatial dimension of the original image. The complete inference workflow, detailing the steps from patch extraction to the final anomaly map generation, is presented in Algorithm 2.

1) Parameter Initialization: All learnable parameters of SM-HAD are initialized as follows. The convolutional layers in the input embedding and output projection [$\mathbf{W}_{\text{conv}}^{\text{in}}$ in (1) and $\mathbf{W}_{\text{conv}}^{\text{out}}$ in (30)] use Kaiming initialization suitable for ReLU-type nonlinearities. Linear projection layers in MVAB and the MLP ($\mathbf{W}_q$, $\mathbf{W}_v$, W_1, W_2) are initialized with Xavier initialization. The complex filters in OSFB ($\mathbf{W}_1^{\mathbb{R}}$, $\mathbf{W}_1^{\mathbb{I}}$, $\mathbf{W}_2^{\mathbb{R}}$, $\mathbf{W}_2^{\mathbb{I}}$) use Xavier normal initialization for both real and imaginary parts. The SSM state matrix $\mathbf{A}^{\text{ssm}}$ and related RMB parameters $\mathbf{W}_{\text{RMB}}$ follow the default Mamba initialization, and the skip parameter $\boldsymbol{\gamma}$ is initialized to ones. All bias terms ($\mathbf{b}_1^{\mathbb{R}}$, $\mathbf{b}_1^{\mathbb{I}}$, $\mathbf{b}_2^{\mathbb{R}}$, $\mathbf{b}_2^{\mathbb{I}}$, $\mathbf{b}_q$, $\mathbf{b}_v$, etc.) are initialized to zero.

Algorithm 2 SM-HAD Inference for Anomaly Detection

Input: Test HSI $\mathbf{X}_{\text{test}} \in \mathbb{R}^{H\times W\times C}$, trained model $\Theta_{\text{SM-HAD}}$
Output: Anomaly map $\mathbf{M} \in \mathbb{R}^{H\times W}$

```
Extract overlapping patches {X̂_i}_{i=1}^N of size P × P × C with stride s
for each patch i = 1, ..., N do
    Forward Pass:
        Apply trained SM-HAD model: X̂_i --Θ_SM-HAD--> X̂_recon,i
    Residual Computation (Eq. (32)):
        R_i = (X̂_i − X̂_recon,i)^2 ∈ R^{P×P×C}
    Residual Diffusion (Eq. (33)-(34)):
        E_i = AvgPool3D(R_i)
        m_i^RD = Σ_{c=1}^C E_i^(c) ∈ R^{P×P}
end for
Anomaly Map Aggregation (Eq. (35)):
        M = F({m_i^RD}_{i=1}^N) ∈ R^{H×W}
return Anomaly map M
```

Note: Higher values in $\mathbf{M}$ indicate anomalies. Threshold at desired false alarm rate.

2) Complementary Design Rationale: MVAB and RMB are intentionally designed as complementary modules: MVAB enforces *explicit* local aggregation through a binary context mask, while RMB provides *efficient* long-range context modeling via a linear-time state–space scan. Although unrestricted (dense) self-attention is, in principle, capable of learning local relations, it often aggregates information from distant patches and thus tends to *over-smooth* subtle, high-frequency anomaly signatures; the masking in MVAB is a structural inductive bias that prevents this mixing and therefore preserves boundary and fine-grained features. Conversely, the Mamba SMM excels at capturing global dependencies with stable training and low complexity (O(L• D) versus O(L^2) for dense attention) but is inherently 1-D/sequential and thus not optimized for the fine 2-D neighborhood structure unless preceded by explicit local mixing. In SM-HAD, we combine a small local-mixing operation (depthwise 1-D/2-D convolution) with the RMB's long-range scan to reap the benefits of both local fidelity and global context while maintaining practical complexity. Ablation (Section V-B) empirically confirms that MVAB improves boundary localization and that RMB supplies complementary global consistency; together they yield the best accuracy/efficiency tradeoff.

IV. Experimental Setup

In this section, we present a comprehensive evaluation of the proposed SM-HAD against 18 SoTA HAD methods across six real-world datasets. The experimental results demonstrate that SM-HAD achieves competitive performance, with superior results on four datasets and effective performance across all evaluated scenarios. All six datasets are sourced from the GT-HAD benchmark [21].

A. Description of Datasets

The Los Angeles-1 and Los Angeles-2 datasets are recorded by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, each consisting of hyperspectral cubes

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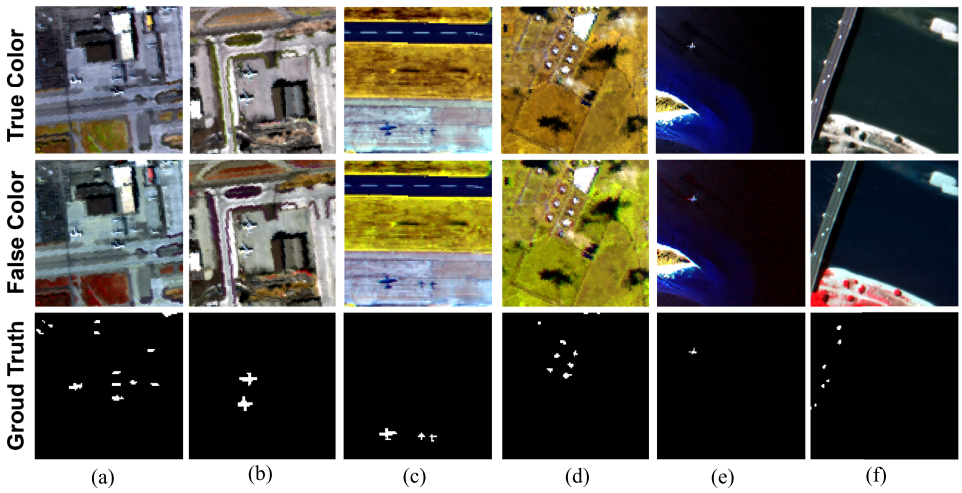


Fig. 3. Visual comparison of true color, false color, and ground truth representations across six benchmark datasets (a) Los Angeles-1, (b) Los Angeles-2, (c) Gulfport, (d) Texas Coast, (e) Cat Island, and (f) Pavia. The first row shows true color RGB composites, the second row displays false color composites highlighting spectral variations, and the third row presents binary ground truth anomaly masks with anomalous pixels shown in white.

with a spatial dimension of 100×100 pixels and 205 spectral bands. These datasets were captured over the Los Angeles region during the early period of AVIRIS campaigns, showcasing the complexities of urban settings with diverse man-made structures and cluttered backgrounds. The spatial resolution of the data is in the order of 7.1 m, and the spectral range is from 400 to 2500 nm, making them suitable for any method that seeks to identify subtle anomaly occurrences in urban scenes [49].

The Gulfport dataset, also acquired by AVIRIS, has 191 spectral bands (excluding the very noisy ones) over a 100×100 spatial grid. The coastal city view with buildings, vegetation, and water bodies is given in a finer spatial resolution of roughly 3.4 m per pixel. Acquired in 2001, this dataset is well known due to the variety of its contents, and it is often employed in the study of cluttered natural-urban transitions [49].

The Texas Coast Dataset has the same specification as AVIRIS; however, it has 204 bands and covers the 100×100 spatial area at a coarser resolution of about 17.2 m. It has a suburban coastal setting of interest for low-resolution anomaly detection. The scene presents challenges due to sparse housing and coastal landforms concerning the mixed-pixel and faint-object-signature problems [49].

The Cat Island dataset features a spatial resolution of 17.2 m and a size of 150×150 pixels with 188 spectral bands. Captured posthurricane by AVIRIS, this dataset comprises mostly natural and coastal features with minimal human-made content. It serves as a useful dataset for studying environmental changes and detecting natural anomalies [49].

The Pavia dataset, acquired by the ROSIS sensor over the University of Pavia, Italy, contains 150×150 spatial pixels and 102 spectral bands in the 430–860 nm range. With a high spatial resolution of 1.3 m/pixel, it is one of the most frequently used datasets for urban classification and anomaly detection due to its spectral clarity and diverse land-cover types, including roads, rooftops, and trees. It was collected in 2003 and remains a benchmark in hyperspectral research [49].

These six benchmark datasets exhibit diverse spectral–spatial characteristics across urban, coastal, and natural environments, as illustrated in Fig. 3 through true color, false color, and ground truth representations.

B. Comparison Methods

In order to thoroughly assess the performance of our proposed SM-HAD model, we performed a comparative analysis with 18 existing HAD methods, categorized into three main groups based on their underlying principles. The statistical-based methods include RX [7], KIFD [29], 2S-GLRT [10], and MsRFQFT [30], which rely on statistical properties and assumptions to identify anomalies. The representation-based methods consist of CRD [31], GTVLRR [36], PTA [50] and PCA-TLRSR [13], which aim to capture lower-dimensional representations or sparse structures for effective anomaly separation. Finally, the deep learning-based methods include LREN [19], Auto-AD [16], GT-HAD [21], PDBSNet [41], DirectNet [42], NL2Net [45], PUNNet [40], BS3LNet [46], SGLNet [48], and HTD-Mamba [47], which leverage neural networks to learn complex and hierarchical spatial–spectral features for improved anomaly detection. This diverse selection of baseline methods ensures a comprehensive and fair evaluation of SM-HAD across various methodological paradigms.

C. Hyperparameter Setting

Among the statistical-based methods, RX does not require any parameter tuning. For KIFD, the number of principal components was varied between 50 and 300 to determine the optimal configuration. The 2S-GLRT and CRD models were evaluated under different window-size settings, where the outer window size (w_{out}) was varied from 5 to 25, and the inner window size (w_{in}) ranged from 3 to 15. Similarly, MsRFQFT was configured with a fixed number of bands ($N_d = 3$), while the width (σ) of its Gaussian kernel was tuned in the range between 0.2 and 10.

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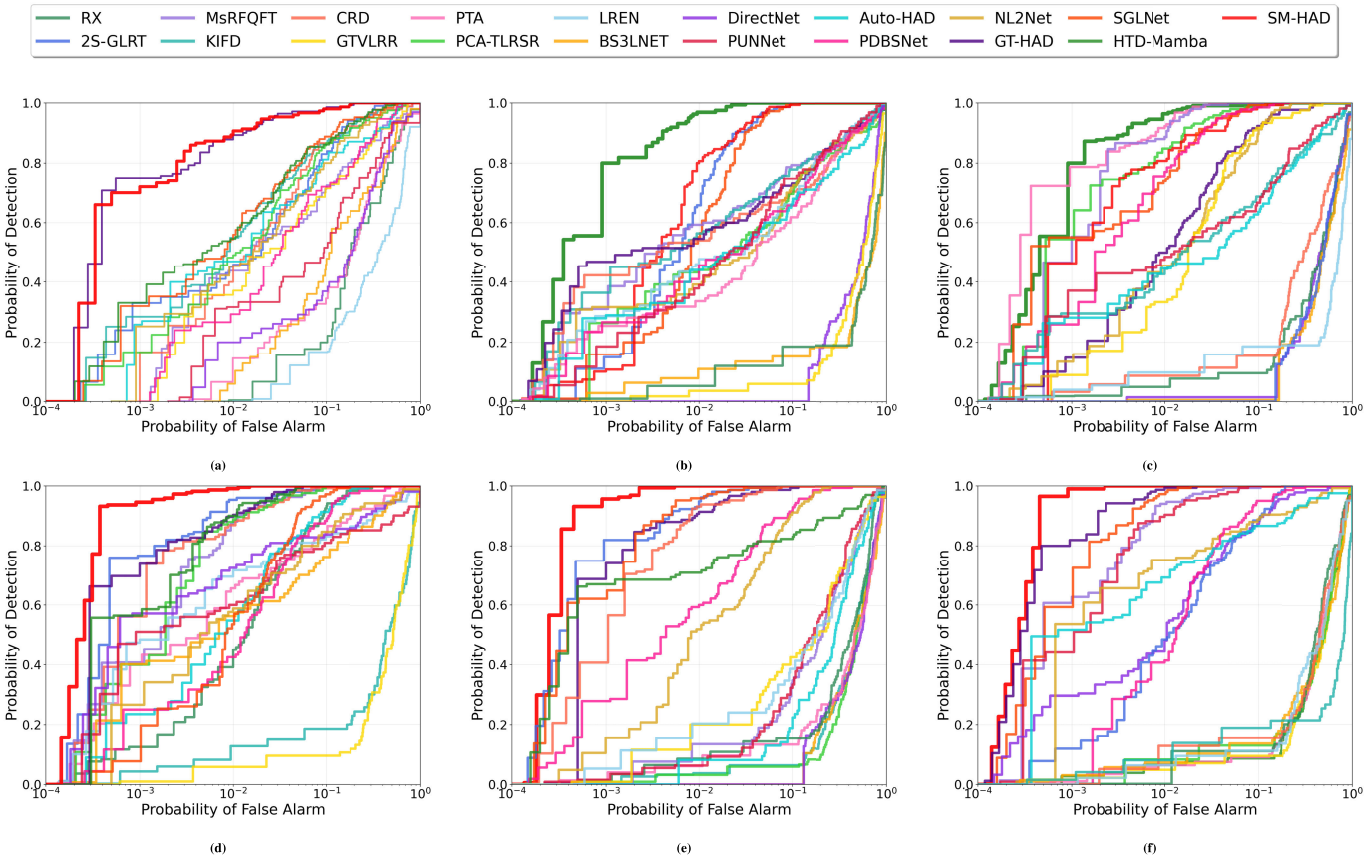


Fig. 4. ROC curves all the methods on the six considered datasets (a) Los-Angeles-1, (b) Los-Angeles-2, (c) Gulfport, (d) Texas-coast, (e) Cat-island, and (f) Pavia.

In the case of representation-based methods, GTVLRR utilized a six-category set, with the top 20 elements of each category forming the dictionary. The model parameters (λ, β, and γ) were set by following the recommendations from the original work [36]. PTA required fixing α, τ, and β to 1, while tuning μ across several logarithmic scales (10^{-4}–10^{4}) and adjusting the low-rank approximation parameter r from 0 to 40. PCA-TLRSR utilized principal components between 1 and 20 with λ and λ' selected from the range 0.001–0.5. For LREN, the regularization parameter λ was tuned from 10^{-6} to 1.

The deep neural network-based models, Auto-AD, GT-HAD, PDBSNet, DirectNet, NL2Net, PUNNet, BS3LNet, SGLNet, HTD-Mamba, and the proposed SM-HAD—were implemented using PyTorch framework on Python 3.10.16. All DNN-based experiments were conducted on an NVIDIA RTX 3090 GPU. The parameters of SM-HAD are optimized using the Adam optimizer [51] with a learning rate of 0.001, a batch size of 64, and trained for 100 epochs. The training convergence of Auto-AD was additionally monitored using a loss variation threshold, chosen between 1.0×10^{-5} and 1.5×10^{-5}. For DirectNet and BS3LNet, overlapping patches were extracted with stride size 5. In contrast, PDBSNet processes the full hyperspectral image directly without patch cropping, utilizing pixel-shuffle downsampling to maintain the blind-spot constraint. HTD-Mamba employed pyramid configurations with multiple resolution at different scales, while NL2Net, PUNNet and SGLNet utilized their respective architectural specifications for nonlocal learning and progressive updates. For all comparative baseline models, we used the original hyperparameter settings as specified in their respective papers to ensure fair evaluation. Since the original HTD-Mamba framework is designed for supervised dense prediction, we adapted it to the unsupervised HAD setting by training it *exclusively on background pixels*. Importantly, no ground-truth anomaly masks were used at any stage. Background pixels were identified using a fully unsupervised two-stage procedure. First, a coarse anomaly map was produced using the classical RX detector, and the top 1% of high-response pixels were removed as potential anomalies. Second, the remaining pixels were refined through median filtering and spatial consistency checking to eliminate isolated outliers. Pixels that satisfied both filtering stages were treated as background and used to train HTD-Mamba in a reconstruction-style manner consistent with the original work. This strategy ensures a fair comparison with other unsupervised methods while allowing HTD-Mamba to learn background spectral–spatial statistics without supervision. All statistical and representation-based methods were implemented in MATLAB 2020a and executed via MATLAB Online. This ensured uniform computational environments and fair comparisons across all models. For SM-HAD, we configured the following hyperparameters: patch size $P = 15$, embedding dimension $D = 96$, Mamba state dimension

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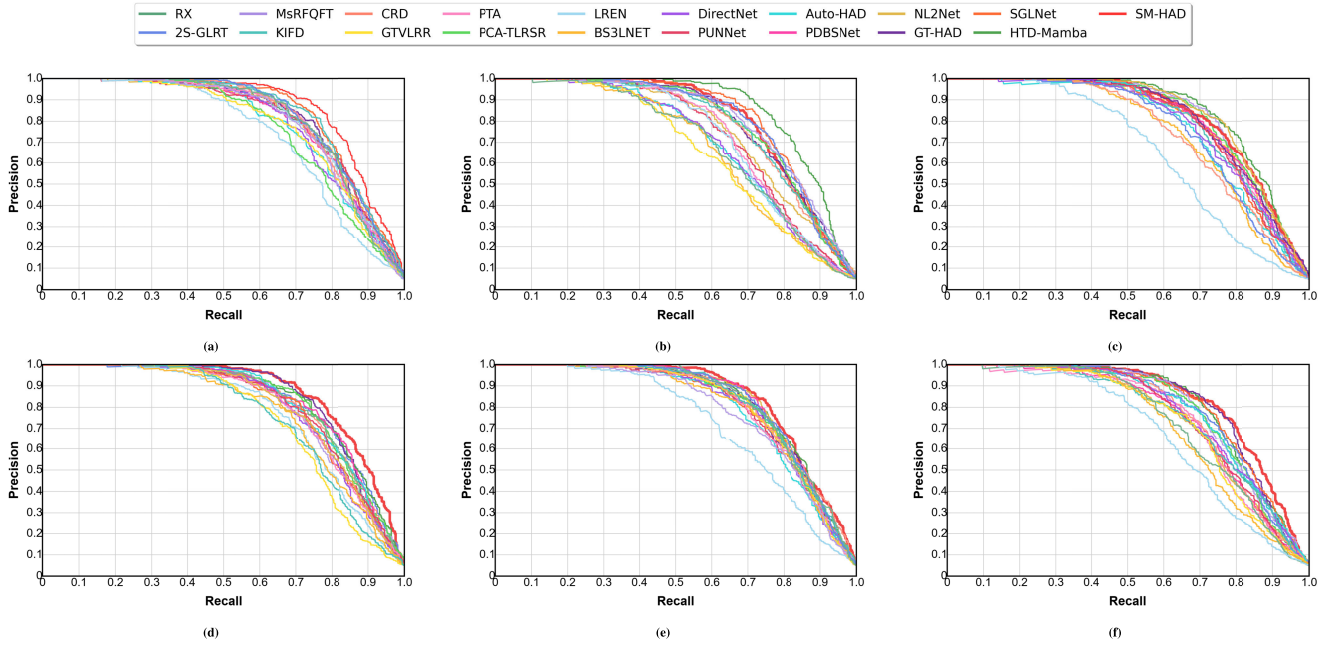


Fig. 5. *P–R* curves of all the methods for the six considered datasets (a) Los-Angeles-1, (b) Los-Angeles-2, (c) Gulfport, (d) Texas-coast, (e) Cat-island, and (f) Pavia.

TABLE I

PERFORMANCE COMPARISON (AUC) OF DIFFERENT HAD METHODS ACROSS SIX BENCHMARK DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD AND THE SECOND-BEST RESULTS ARE UNDERLINED

Method Type	Method	LA-1	LA-2	Gulfport	Texas Coast	Cat Island	Pavia	Avg AUC
Statistical Models	RX	0.8221	0.8404	0.9526	0.9907	0.9807	0.9538	0.9234
	2S-GLRT	0.9236	0.9873	0.9183	0.9970	0.9995	0.9867	0.9687
	MsRFQFT	0.9114	0.9798	0.9945	0.9953	0.9798	0.9984	0.9765
	KIFD	0.9359	0.9775	0.9728	0.9303	0.9900	0.8634	0.9450
Representation-based	CRD	0.9212	0.9681	0.9445	0.9940	0.9946	0.9407	0.9605
	GTVLRR	0.9004	0.8840	0.9874	0.9478	0.9694	0.9747	0.9440
	PTA	0.8809	0.9104	0.9946	0.9757	0.9831	0.9749	0.9533
	PCA-TLRSR	0.9455	0.9664	0.9930	0.9923	0.9854	0.9635	0.9744
Deep Learning	LREN	0.7327	0.9134	0.8293	0.9783	0.9343	0.8985	0.8811
	BS3LNET	0.8550	0.8424	0.9467	0.9583	0.9847	0.9593	0.9244
	DirectNet	0.8917	0.8912	0.9598	0.9896	0.9841	0.9878	0.9507
	PUNNet	0.8755	0.9056	0.9652	0.9819	0.9792	0.9914	0.9498
	Auto-HAD	0.9207	0.9063	0.9672	0.9909	0.9783	0.9807	0.9574
	PDBSNet	0.9005	0.9100	0.9905	0.9913	0.9926	0.9887	0.9623
	NL2Net	0.9235	0.9209	0.9876	0.9702	0.9914	0.9843	0.9630
	GT-HAD	0.9515	0.9666	0.9888	0.9969	0.9984	0.9993	0.9836
	HTD-Mamba	0.9465	**0.9942**	**0.9978**	0.9961	0.9907	0.9800	0.9842
	SGLNet	0.9377	0.9844	0.9915	0.9905	0.9952	0.9950	0.9824
Proposed (Ours)	**SM-HAD**	**0.9793**	0.9833	0.9929	**0.9982**	**0.9996**	**0.9994**	**0.9921**

$d_s = 64$, soft-shrinkage threshold $\lambda = 0.01$, learning rate $\eta = 0.001$ and batch size $B = 64$. These values were selected through preliminary experiments. Smaller patches ($P < 10$) fail to capture sufficient spatial context, while larger patches ($P > 20$) increase computational overhead. Testing embedding dimensions $D \in \{64, 96, 128, 192\}$ showed that $D = 96$ provides the best tradeoff between representational capacity and efficiency. For state dimension $d_s \in \{16, 32, 64, 128\}$, we observed that smaller values ($d_s \leq 32$) led to unstable training and inconsistent detection performance across runs, while $d_s = 64$ consistently achieved the highest AUC with stable convergence.

D. Evaluation Metrics

We use a comprehensive set of metrics to thoroughly evaluate anomaly detection algorithms. ROC curves (Fig. 4) plot the relationship between true positive and false positive rates, allowing us to compare how well each method distinguishes anomalies from normal data across thresholds.

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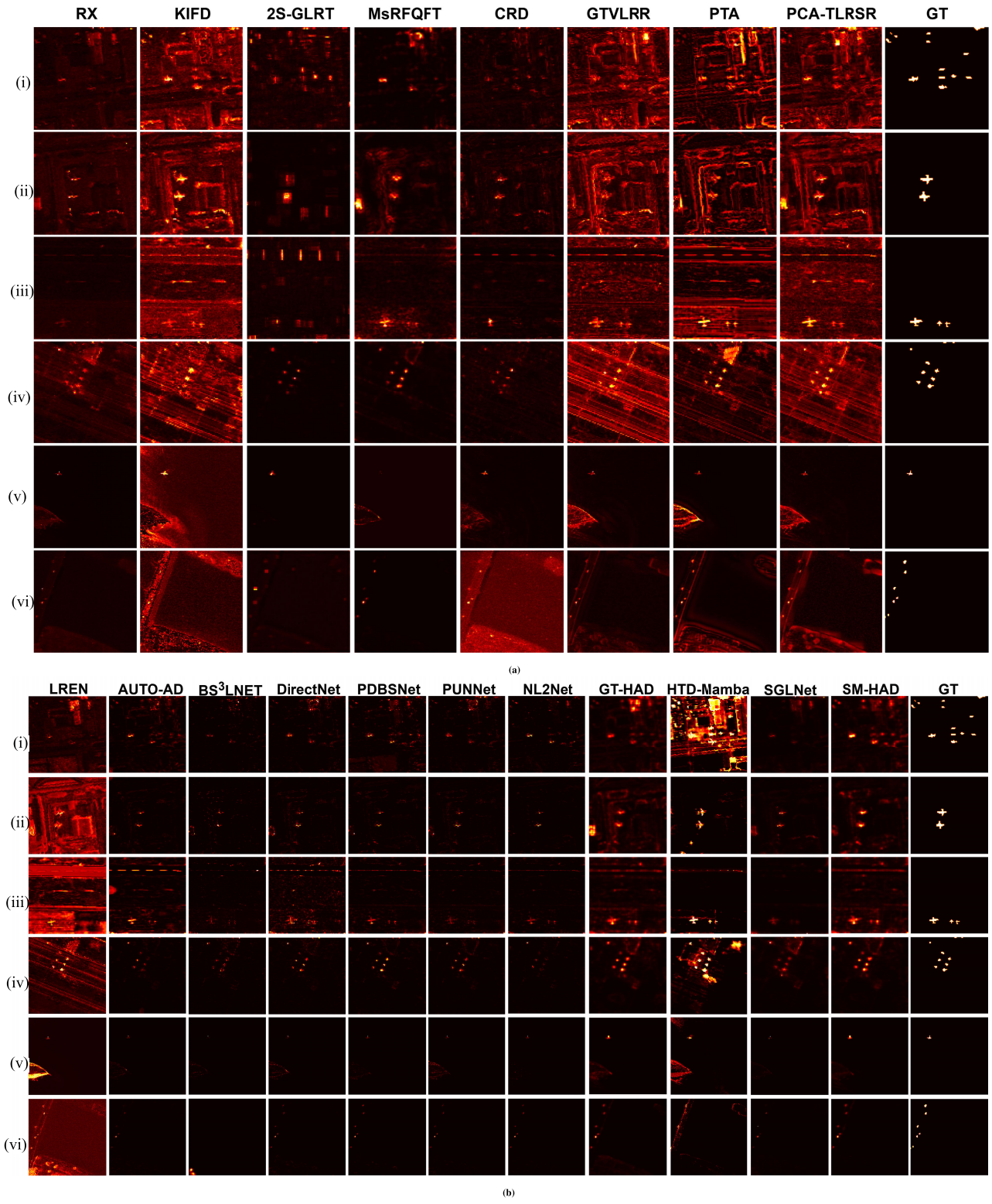


Fig. 6. Comparison of anomaly maps using (a) traditional methods and (b) deep learning models on six considered datasets (i) Los Angeles-1, (ii) Los Angeles-2, (iii) Gulfport, (iv) Texas Coast, (v) Cat Island, and (vi) Pavia.

The AUC (Table I) summarizes this performance numerically, with higher values indicating better overall discrimination. 3-D ROC curves provide an extended view by incorporating additional variables or conditions. Precision–Recall (P–R) curves (Fig. 5) point out the tradeoff between precision and recall, which is particularly valuable in scenarios with rare anomalies, highlighting how robust each detector is to class imbalance. We also include anomaly maps and binary ground truth images (Fig. 6) to qualitatively assess how accurately and consistently anomalies are localized and backgrounds are

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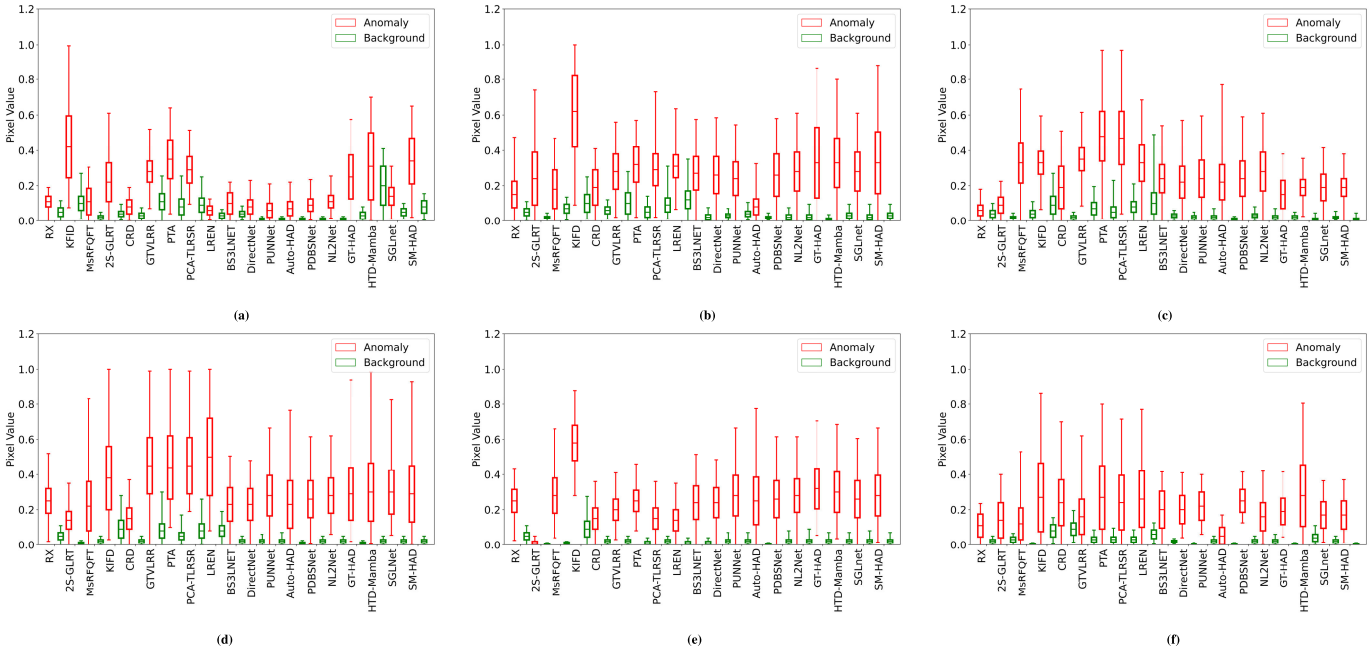


Fig. 7. Separability maps obtained of all the methods for the six datasets (a) Los-Angeles-1, (b) Los-Angeles-2, (c) Gulfport, (d) Texas-coast, (e) Cat-island, and (f) Pavia.

suppressed. Finally, separability maps, shown as box plots (Fig. 7), depict the distribution of scores for anomalies and backgrounds greater separation indicates more reliable detection. Together, these metrics provide comprehensive evaluation of both quantitative accuracy and qualitative detection characteristics, enabling objective assessment of each method's performance.

V. Results and Analysis

In this section, we first present quantitative performance comparisons of SM-HAD against eighteen SoTA methods on six benchmark datasets Table I. We then analyze threshold-based detection characteristics via ROC and *P–R* curves, followed by qualitative visualizations—including separability box plots and anomaly maps—to illustrate anomaly localization and robustness across diverse scenarios.

A. Comparison of Different Methods

To provide focused insight into deep learning model effectiveness, we compare SM-HAD with the five highest-scoring deep learning-based methods for each dataset. The complete set of results, including traditional models, is available in Table I.

1) Los Angeles-1: SM-HAD achieves an AUC of 0.9793, outperforming GT-HAD (0.9515), HTD-Mamba (0.9465), SGLNet(0.9377), and NL2Net (0.9235). The improvements over GT-HAD and HTD-Mamba are approximately 2.9% and 3.3%, respectively, demonstrating SM-HAD's strong capability to separate anomalies in complex urban–coastal environments.

2) Los Angeles-2: HTD-Mamba attains the highest AUC (0.9942), followed by (0.9844) and SM-HAD (0.9833). Although SM-HAD is 1.1% lower than HTD-Mamba and slightly below SGLNet, it surpasses GT-HAD (0.9666) and NL2Net (0.9209). These results indicate that SM-HAD maintains competitive performance even when the anomaly distribution differs substantially from other datasets.

3) Gulfport: HTD-Mamba yields the best AUC at 0.9978, with SM-HAD closely behind at 0.9929, followed by SGLNet (0.9915) and PDBSNet (0.9905). SM-HAD exceeds GT-HAD (0.9888) by 0.4% and surpasses SGLNet. This strong performance in vegetation-dominated regions highlights SM-HAD's robustness across complex background conditions.

4) Texas Coast: SM-HAD achieves the highest AUC of 0.9982, outperforming GT-HAD (0.9969), HTD-Mamba (0.9961), PDBSNet (0.9913), and SGLNet (0.9905). Although the margins over GT-HAD (0.13%) and HTD-Mamba (0.21%) are modest, they show that SM-HAD handles mixed land–water spectral signatures with high precision.

5) Cat Island: SM-HAD obtains the best AUC at 0.9996, exceeding GT-HAD (0.9984) and SGLNet (0.9952). PDBSNet (0.9926) and NL2Net (0.9914) perform notably lower. The 0.12% improvement over GT-HAD in a sparsely populated anomaly environment underscores the model's fine spectral discrimination capability.

6) Pavia: SM-HAD reaches an AUC of 0.9994, slightly outperforming GT-HAD (0.9993) and substantially surpassing SGLNet (0.9950) and HTD-Mamba (0.9800). This consistent superiority in an urban scene demonstrates SM-HAD's ability to capture subtle spectral variations in dense and heterogeneous environments.

In summary, SM-HAD achieves competitive results among deep learning-based methods, ranking within the top three on all datasets, with percentage differences ranging from 0.01% to 2.9% over competitors. Even in LA-2 and Gulfport, where HTD-Mamba attains higher AUCs, SM-HAD demonstrates

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TABLE II
ABLATION STUDY PERFORMANCE ACROSS THE SIX CONSIDERED DATASETS. VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION. TOP SCORES ARE HIGHLIGHTED IN BOLD

Method	Los-Angeles-1	Los-Angeles-2	Gulfport	Texas-Coast	Cat-Island	Pavia
Base	0.8733±5.15e-2	0.9592±6.98e-4	0.9783±1.18e-2	0.9857±8.09e-4	0.9858±5.55e-3	0.9378±3.07e-2
Base + MVAB	0.9476±1.51e-2	0.9628±4.03e-3	0.9885±5.02e-3	0.9935±3.21e-3	0.9979±5.10e-4	0.9987±3.84e-5
Base + RMB	0.8338±5.93e-2	0.9667±5.86e-3	0.9742±1.83e-2	0.9855±2.06e-3	0.9877±6.07e-3	0.9668±5.14e-3
Base + OSFB	0.9050±1.50e-2	0.9778±5.25e-4	0.9834±1.13e-2	0.9890±1.84e-3	0.9858±3.15e-3	0.9794±8.74e-3
Base + OSFB + RMB	0.9023±2.40e-3	0.9773±4.68e-4	0.9891±5.88e-3	0.9907±2.05e-3	0.9854±2.68e-3	0.9832±5.62e-3
Base + MVAB + RMB	0.9536±1.07e-2	0.9667±3.59e-3	0.9918±4.05e-3	0.9951±2.09e-3	0.9989±3.23e-4	0.9994±2.78e-5
Base + MVAB + OSFB	0.9790±1.58e-3	0.9833±7.99e-4	0.9917±1.67e-3	0.9973±1.01e-3	0.9996±5.84e-6	0.9995±2.05e-5
Base + RMB + OSFB + MVAB	**0.9794±1.97e-3**	**0.9833±1.25e-3**	**0.9929±2.40e-3**	**0.9983±1.37e-4**	**0.9996±2.92e-6**	**0.9995±1.85e-5**

clearer and more precise anomaly localization, as shown in Fig. 6. Methods that score higher by AUC in those cases often detect less distinct anomaly regions visually. These observations indicate that SM-HAD is not only numerically competitive but also visually effective, supporting its practical relevance for HAD.

The ROC curves in Fig. 4 consistently demonstrate SM-HAD's discrimination capability across all six datasets. In each plot, SM-HAD's curve (red) is closest to the upper-left corner, indicating the highest true-positive rate at a given false-positive rate—especially excelling in the critical ultralow region of 10^{-4} to 10^{-2}. On LA-1, Texas Coast, Cat Island, and Pavia, SM-HAD's ROC trace dominates the envelope, achieving steeper rises and earlier plateaus compared to all baselines. On LA-2 and Gulfport, although HTD-Mamba leads, SM-HAD remains among the top performers with minimal degradation. The tight clustering of SM-HAD's curve near the ideal point in all scenarios confirms its robustness in detection while minimizing false alarms, validating its practical effectiveness for real-world HAD.

The *P–R* curves in Fig. 5 further validate SM-HAD's effectiveness under class-imbalanced conditions. SM-HAD maintains consistent precision across increasing recall values, significantly outperforming baseline methods with a larger area under the *P–R* curve. On challenging datasets like LA-1 and Gulfport, SM-HAD exhibits stable precision even at high recall, while competing methods show rapid precision degradation. This performance confirms SM-HAD's robustness against false positives when anomalies represent only a small fraction of total pixels.

The separability maps in Fig. 7 reveal SM-HAD's discriminative capability through clear separation between anomaly and background score distributions. SM-HAD tends to produce minimal overlap between interquartile ranges with compact score clustering across all datasets. The box plots show that SM-HAD assigns higher scores to true anomalies while background scores remain lower. In contrast, competing methods exhibit substantial distribution overlap, particularly on complex datasets like LA-1 and Gulfport, which may suggest comparatively reduced discriminative effectiveness.

The anomaly maps in Fig. 6 demonstrate SM-HAD's superior spatial localization accuracy. SM-HAD produces spatially coherent detections with sharp boundaries that closely match ground truth patterns while effectively suppressing background false alarms. Traditional statistical methods like RX show significant background clutter, while representation-based approaches such as CRD exhibit fragmented detections. The visual comparison confirms SM-HAD's spatial consistency and accurate boundary delineation across diverse environmental conditions.

B. Ablation Study

In this section, we present a comprehensive ablation study conducted across all six benchmark datasets to strictly validate the contribution of each proposed component in SM-HAD. We analyze the model from two perspectives: 1) an incremental integration analysis (Table II and Fig. 8) to demonstrate the synergistic effect of the core modules and 2) a component removal analysis (Table III) to verify the necessity of specific internal mechanisms. All experiments were repeated three times, and mean results with standard deviations are reported.

1) Impact of Core Modules (Incremental Analysis): Table II details the performance evolution as we progressively integrate the MVAB, RMB, and the OSFB into the baseline architecture.

a) Necessity of local context (MVAB versus RMB): A critical observation arises when comparing Base + MVAB against Base + RMB. Incorporating MVAB yields a substantial performance boost (e.g., increasing AUC from 0.8733 to 0.9476 on Los Angeles-1), confirming that preserving local 2-D spatial neighborhoods is essential for detecting anomalies. Conversely, utilizing RMB alone (Base + RMB) leads to a performance drop on several datasets compared to the baseline. This empirically validates our design rationale: while Mamba is efficient for long-range dependencies, its 1-D sequential scanning can disrupt local 2-D spatial structures if not preceded by explicit local feature aggregation. The combination of Base + MVAB + RMB recovers this performance, proving that MVAB effectively primes the features with local context, allowing RMB to subsequently capture global dependencies without compromising local fidelity.

b) Role of spectral modeling (OSFB): The addition of OSFB (Base + OSFB) consistently improves performance across all datasets (e.g., + 3.1% on Los Angeles-1 over Base). This confirms that projecting features into an orthogonal frequency domain reduces spectral redundancy and enhances

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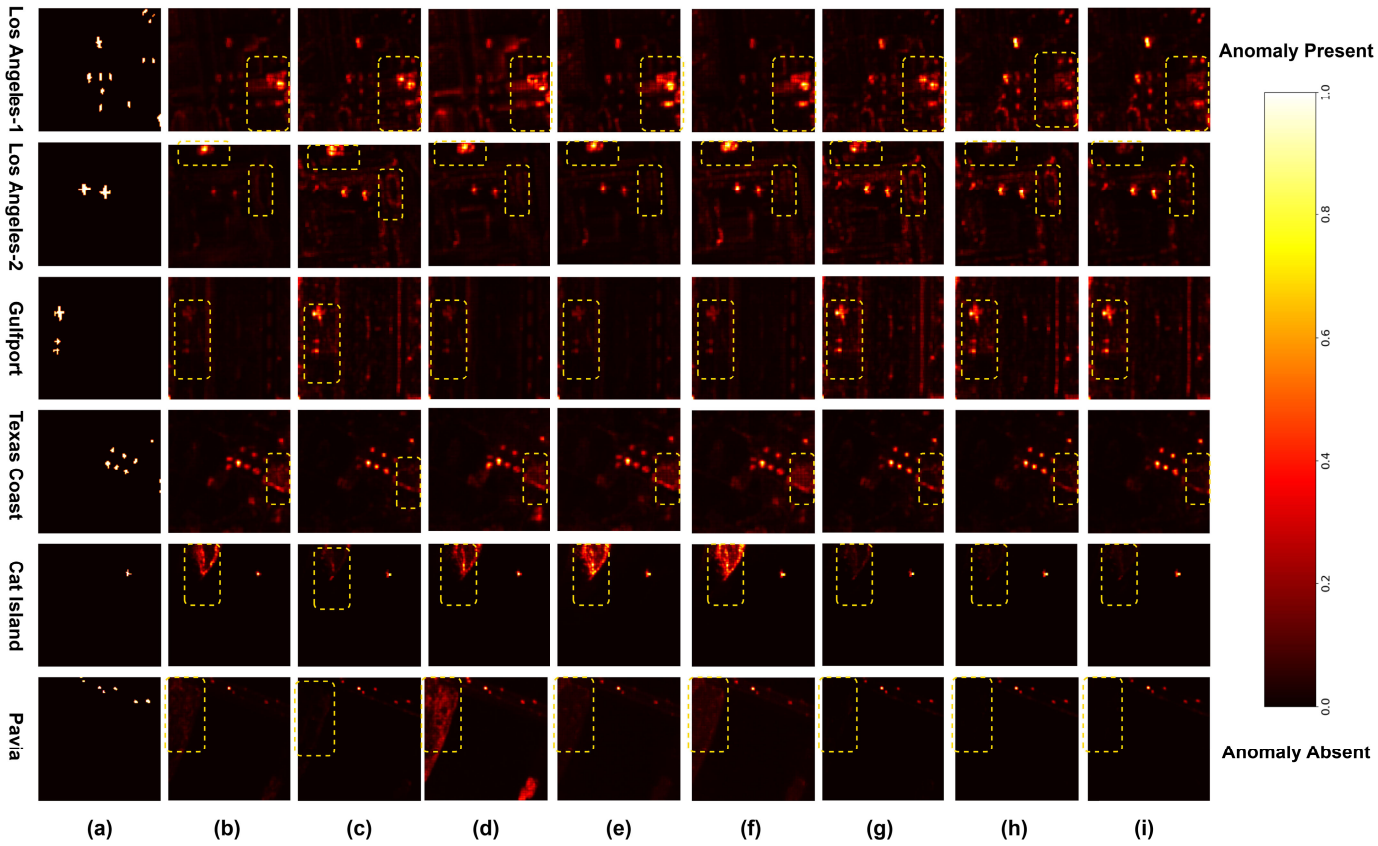


Fig. 8. Qualitative ablation results across the six datasets. (a) Ground truth. (b) Base. (c) Base + MVAB. (d) Base + RMB. (e) Base + OSFB. (f) Base + OSFB + RMB. (g) Base + MVAB + RMB. (h) Base + MVAB + OSFB. (i) Full SM-HAD.

TABLE III

ABLATION STUDY INVESTIGATING THE IMPACT OF REMOVING SUBCOMPONENTS FROM THE PROPOSED ARCHITECTURE. VALUES ARE REPORTED AS MEAN ± STANDARD DEVIATION. THE PERFORMANCE DROP IN EACH ROW VALIDATES THE CONTRIBUTION OF THE REMOVED MODULE

Model Variant	Los-Angeles-1	Los-Angeles-2	Gulfport	Texas-Coast	Cat-Island	Pavia
Proposed (Full Model)	**0.9794±1.97e-3**	**0.9833±1.25e-3**	**0.9929±2.40e-3**	**0.9983±1.37e-4**	**0.9996±2.92e-6**	**0.9995±1.85e-5**
w/o Residual Diffusion	0.9685±1.45e-2	0.9740±3.20e-3	0.9880±3.50e-3	0.9925±2.10e-3	0.9975±4.50e-4	0.9982±3.80e-5
w/o FFT in OSFB	0.9615±1.20e-2	0.9680±3.80e-3	0.9890±4.10e-3	0.9940±2.50e-3	0.9980±4.20e-4	0.9988±3.50e-5
w/o Masked Attention	0.9650±1.10e-2	0.9710±2.90e-3	0.9905±3.20e-3	0.9955±1.80e-3	0.9985±3.10e-4	0.9990±2.90e-5
w/o Soft-Shrinkage	0.9695±1.55e-3	0.9755±1.90e-3	0.9910±2.25e-3	0.9962±1.35e-3	0.9988±1.80e-5	0.9991±2.40e-5

the discriminability of anomaly signatures. The full SM-HAD model (Base + RMB + OSFB + MVAB) achieves the highest scores across all metrics, demonstrating that spectral (OSFB), local-spatial (MVAB), and global-sequential (RMB) information are complementary and mutually reinforcing.

2) Critical Subcomponent Analysis (Component Removal): To further justify the internal design choices of our blocks, we investigate the impact of removing specific subcomponents in Table III.

a) Impact of frequency domain processing (w/o FFT): Removing the FFT operation from the OSFB module forces the network to rely solely on spatial domain convolutions. This results in a notable accuracy drop (e.g., AUC declines from 0.9794 to 0.9615 on Los Angeles-1), indicating that the global spectral dependencies captured via the frequency domain are critical for distinguishing subtle spectral anomalies that are otherwise indistinguishable in the spatial domain.

b) Impact of masking strategy (w/o masked attention): When the masking mechanism is removed from the attention block (allowing dense global attention), the performance degrades consistently. This degradation supports the hypothesis that unrestricted attention leads to "oversmoothing," where the distinct spectral signatures of small anomalies are averaged out by the background. The binary masking successfully acts as a structural inductive bias, preserving high-frequency anomaly details.

c) Impact of sparsity (w/o soft-shrinkage): Removing the soft-shrinkage thresholding from the OSFB reduces performance. This suggests that the shrinkage operation effectively suppresses noise and uninformative frequency components, acting as a spectral denoising filter that refines the feature representation.

d) Impact of RD (w/o RD): As observed in Table III, removing RD results in a performance drop across all datasets

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TABLE IV
FLOPS, PARAMETERS, AND RUNTIME OF SM-HAD VERSUS STATE-OF-ART METHODS ON THE LOS ANGELES-1 DATASET

Models	LREN	Auto-AD	BS3LNet	DirectNet	PDBSNet	PUNNet	NL2Net	HTD-Mamba	GT-HAD	SGLNet	SM-HAD
FLOPs	11.97G	0.19G	1.65G	1.65G	3.23G	1.31G	3.13G	**0.13G**	2.64G	5.46G	1.46G
Params.	3.25M	0.86M	0.40M	0.40M	0.79M	1.19M	0.76M	0.32M	**0.26M**	0.55M	0.28M
Runtime	30.55s	161.85s	128.55s	130s	144.3s	140.76s	176.2s	549.02s	29.54s	54s	**25.65s**

(e.g., AUC decreases to 0.9685 on Los Angeles-1 and 0.9880 on Gulfport). This confirms that spatially diffusing the residuals effectively highlights anomaly boundaries and suppresses isolated noise, thereby refining the final detection map.

C. Model Complexity

We evaluated the computational efficiency of SM-HAD in terms of FLOPs, Parameters (Params), and Runtime against other deep learning methods across ten DNN-based approaches on the Los Angeles-1 dataset. As shown in Table IV, SM-HAD is relatively efficient with 0.28 M parameters and 1.46 G FLOPs. This is significantly lower than resource-intensive models like LREN (3.25 M parameters, 11.97 G FLOPs) and PDBSNet (0.79 M parameters, 3.23 G FLOPs). Similarly, SGLNet incurs higher computational costs with 0.55 M parameters and 5.46 G FLOPs. Although HTD-Mamba requires the lowest FLOPs at 0.13 G, it suffers from the longest runtime of 549.02 s, making it impractical for real-time applications. In terms of Runtime, SM-HAD achieves a runtime of 25.65 s among all evaluated methods, outperforming GT-HAD (29.54 s), SGLNet (54.00 s), and Auto-AD (161.85 s). Despite GT-HAD possessing a slightly lower parameter count (0.26 M), SM-HAD offers a more favorable compromise between runtime and parameter efficiency. These results suggest that SM-HAD is an efficient and resource-conscious approach for HAD, with practical potential for real-time applications due to its relatively low computational demands.

VI. CONCLUSION

In this article, we propose SM-HAD, a novel framework that successfully integrates frequency-domain analysis, attention mechanisms, and state–space modeling to address some of the core challenges in HAD. The three key components—OSFB for global spectral modeling, MVAB for spatial–spectral attention, and RMB for efficient dependency encoding—work synergistically to achieve very good detection performance while maintaining computational practicality for real-world deployment. Our comprehensive experimental validation confirms the effectiveness of combining these complementary techniques, establishing a new approach for balancing accuracy and efficiency in HAD tasks. The framework's ability to maintain high performance with minimal computational overhead makes it particularly suitable for practical applications where resource constraints are critical. This work opens new research directions in hyperspectral processing by demonstrating the potential of hybrid architectures that leverage both spectral and spatial information through multiple representation domains. The successful integration of the Mamba-based SMM with frequency-domain processing represents a significant advancement in the field.

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